Convergence and Obstructions for the Implemented PsiLogic Optimizer

An implementation-grounded theory with explicit scope and proof audit

Research appendix for the PsiLogic working tree

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Abstract

We reconstruct PsiLogic from the local executable Python and Triton source, including current-gradient cancellation, normalized norm statistics, bias correction, preprocessing, and two distinct decay mechanisms. For its real-arithmetic Adam configuration without gradient preprocessing or parameter decay, we prove a pathwise finite-horizon stationarity bound with the full implemented detector and both moments. The bound gives $O(T^{-1})$ average squared gradient for deterministic gradients at a sufficiently small constant step, and quantitative guarantees with vanishing or sufficiently small relative oracle error. It does not establish convergence under arbitrary persistent stochastic noise. A simplified predictable-multiplier method admits the usual $O(T^{-1/2})$ stochastic bound; its predictability is not a property of the implemented detector. Explicit examples demonstrate a fixed-step two-cycle, current-gradient anti-alignment, and permanent nonstationary stalling under gradient centralization. We bound detector states and cancellation, disclose finite-precision and backend limitations, and provide source traces, independent recurrence tests, and a separate formal-verification status. No unconditional guarantee for the default configuration or superiority over AdamW is claimed.

1 Introduction and contributions

Adaptive normalization and momentum are useful algorithmic mechanisms, but neither positive step multipliers nor small empirical loss establish a convergence theorem. PsiLogic adds a multiplier computed from the same gradient used in its update. This creates a conditional dependence that must be retained in the analysis. The present object of study is the local package at revision `b4917163d63d8a5cb2960e29af4b1bccdbf5798d`, with its working-tree state recorded in `source_snapshot.json`. Existing papers, documentation, and release equations were not used as the algorithm specification.

The main contributions are: (i) exact equations and branch coverage; (ii) uniform real-arithmetic bounds on the implemented detector; (iii) a pathwise error-energy theorem for bias-corrected momentum with arbitrary, possibly current-gradient-dependent, uniformly positive diagonal scaling; (iv) its specialization to an actual PsiLogic configuration, with explicit constants; and (v) counterexamples defining the limits of stronger claims. The main theorem does not assume monotonic second moments, independence of trust and gradients, or the convergence of classical Adam. Its sufficient step bound can be extremely conservative when epsilon is small. It is a rigorous existence guarantee, not a recommendation for training hyperparameters.

2 Exact algorithm specification

2.1 Blocks, clocks, defaults, and preprocessing

Partition $x \in \mathbb{R}^d$ into parameter tensors $x^{(b)} \in \mathbb{R}^{d_b}$; tensor shape is retained for centralization. A block with `grad is None` receives no update, no decay, and no local-clock increment. In this section $k \geq 1$ is a block's local active-update count; x denotes its pre-step value. Group options may differ. Unless otherwise specified, all formulas are exact-real interpretations, not statements about floating-point evaluation.

The constructor defaults are

$$\eta = 10^{-3}, \quad (\beta_1, \beta_2) = (0.9, 0.999), \quad \lambda = 10^{-4}, \quad \gamma = 0.05, \quad p_{\text{ext}} = 1,$$
$$q = 0, \quad \epsilon = 10^{-8}, \quad c_{\max} = 0.05, \quad \tau = 0.5, \quad \tau_{\text{scale}} = 2, \quad H = 0, \quad W_{\text{cfg}} = -1.$$

AGC, centralization, Lion, auto-gamma, and DDP chaos synchronization default to false (AGC coefficient zero); adaptive thresholding, foreach, and fused CUDA selection default to true. Task presets and parameter-group builders can change these choices. No global norm clipping is implemented by this optimizer; external training-loop clipping is outside its update path.

Given raw supplied gradient z, AGC with coefficient $a_{\text{gc}} > 0$ computes

$$\chi = \min\left\{1, \frac{a_{\text{gc}} \max(\|x\|, 10^{-3})}{\max(\|z\|, 10^{-6})}\right\}, \qquad r = \chi z. \tag{1}$$

With AGC disabled, $r = z$. Centralization gives $g = Pr$, where P subtracts the mean over every axis except the leading axis if tensor rank exceeds one; otherwise $P = I$. All moments and detector norms use g. Quantum decay uses r, specifically *after AGC and before centralization*. The norm in (1) is a whole-tensor Euclidean norm, not a per-row norm.

2.2 Detector, schedules, and cancellation

The initialized state is $m_0 = v_0 = 0$, $A_0 = F_0 = S_0 = 0$, stored soft chaos zero, and local count zero. Moments are block vectors; A, F, S and soft chaos are one-element tensors. Define $R_k = \|g_k\| / \sqrt{\max(d_b, 1)}$. At the first active step,

$$A_1 = R_1, \qquad F_1 = S_1 = 1, \tag{2}$$

even when $g_1 = 0$. At later steps,

$$A_k = .99A_{k-1} + .01R_k, \quad Z_k = \frac{R_k}{A_k + \epsilon}, \quad F_k = .9F_{k-1} + .1Z_k, \quad S_k = .99S_{k-1} + .01Z_k. \tag{3}$$

If enabled and a process group exists, DDP replaces F_k, S_k by their arithmetic means across ranks. It does not average A_k, local disagreement, or local moment state. Consistent collective ordering is required.

Disagreement reads the *old* momentum:

$$D_k = \begin{cases} 0, & k = 1, \\ \frac{1}{2}\left[1 - \text{clip}\left(\dfrac{\langle g_k, m_{k-1}\rangle}{\|g_k\| \, \|m_{k-1}\| + \epsilon}, -1, 1\right)\right], & k > 1. \end{cases} \tag{4}$$

It is computed only when cancellation can be active. Write $\sigma(u) = 1/(1+e^{-u})$. The excess and soft chaos are

$$E_k = \begin{cases} \sigma\left(\dfrac{4}{\max(\tau_{\text{scale}}-1,.25)}\left(\dfrac{F_k}{S_k+\epsilon}-1\right)\right), & \text{adaptive,} \\ \sigma\left(4\left(\dfrac{S_k}{\max(\tau,\epsilon)}-1\right)\right), & \text{otherwise,} \end{cases} \qquad s_k = \tanh(S_k)\frac{E_k+D_k}{2}. \tag{5}$$

The detector is continuous on its positive-denominator domain, including clamping and the sigmoid; it has no hard spike indicator. The first-step seed and schedule branches are discrete decisions.

For horizon $H = \texttt{gamma_T_max}$ set

$$C_k = \begin{cases} (1+\cos(\pi\min(k/H,1)))/2, & H>0, \\ 1, & H\le 0, \end{cases} \qquad \gamma_k^{\text{sched}} = \gamma C_k, \quad q_k^{\text{sched}} = qC_k.$$

For nonnegative configured warmup W_{cfg} use $W = W_{\text{cfg}}$; otherwise $W = \max(500, \lfloor H/20 \rfloor)$ if $H > 0$, and $W = 200$ otherwise. Then

$$w_k = \begin{cases} 0, & k \le W, \\ \min(1,(k-W)/\max(1,\lfloor W/4\rfloor)), & k > W. \end{cases}$$

Auto-gamma, if enabled, multiplies γ_k^{sched} by $\max(S_k/.1,.1)$ when $k>1$ and $S_k<.1$, and by one otherwise. Denote the result by γ_k^{eff}. This uses the current, possibly synchronized, slow state. In the active branch $w_k > 0$, $\gamma_k^{\text{eff}} > 0$, set

$$c_* = \min(c_{\max},.99), \qquad u_k = 1-\min\{s_k\gamma_k^{\text{eff}}p_{\text{ext}}w_k, c_*\}, \qquad Q_k = q_k^{\text{sched}}w_k(1-s_k). \tag{6}$$

Here Q_k is used only if $q_k^{\text{sched}} > 0$; otherwise $Q_k = 0$. In an inactive branch $u_k = 1, Q_k = 0$, and the stored soft chaos is set to zero. Thus $\gamma = 0$ also disables quantum decay, even when $q > 0$.

2.3 Adam and Lion are different recurrences

In Adam mode,

$$m_k = \beta_1 m_{k-1} + (1-\beta_1)g_k, \qquad v_k = \beta_2 v_{k-1} + (1-\beta_2)g_k^{\odot 2}, \tag{7}$$

$$x^+ = (1-\eta_k\lambda)(1-\eta_k Q_k\tanh|r_k|)\odot x - \eta_k\frac{\sqrt{1-\beta_2^k}}{1-\beta_1^k}\frac{u_k m_k}{\sqrt{v_k}+\epsilon}. \tag{8}$$

All vector products, square roots, and quotients in (8) are coordinatewise. Weight decay precedes quantum decay; their product contains a cross term and neither is scaled by trust. With $\widehat{m}_k = m_k/(1-\beta_1^k)$ and $\widehat{v}_k = v_k/(1-\beta_2^k)$, the adaptive term is

$$-\eta_k u_k\frac{\widehat{m}_k}{\sqrt{\widehat{v}_k}+\epsilon/\sqrt{1-\beta_2^k}}. \tag{9}$$

This epsilon placement differs from the commonly written Adam update with constant epsilon in the bias-corrected denominator.

In Lion mode the recurrence is instead

$$\begin{aligned} \ell_k &= \operatorname{sign}\big(\beta_1 m_{k-1} + (1-\beta_1)g_k\big), \\ x^+ &= (1-\eta_k\lambda)(1-\eta_k Q_k\tanh|r_k|)\odot x - \eta_k u_k \ell_k, \\ m_k &= \beta_2 m_{k-1} + (1-\beta_2)g_k, \end{aligned} \tag{10}$$

with $\operatorname{sign}(0) = 0$, no bias correction, and unchanged variance state. Our main convergence theorem concerns (8), not (10).

3 Problem formulation and explicit assumptions

For a positive integer horizon T, use global indices $t = 1, \dots, T$, with x_t the point before update t and $h_t = \nabla f(x_t)$. Let $\mathcal{F}_{t-1}$ contain x_1, all gradients observed before update t, pre-step optimizer states, and any history-dependent group settings. Thus $x_t, h_t, m_{t-1}, v_{t-1}, A_{t-1}, F_{t-1}, S_{t-1}$ are $\mathcal{F}_{t-1}$-measurable. With the current gradient included they generate $\mathcal{F}_t$. Preprocessed gradients, disagreement, updated detector states, trust, updated moments, and x_{t+1} are $\mathcal{F}_t$-measurable, generally not $\mathcal{F}_{t-1}$-measurable. In a distributed formulation the filtration must include the ranks' joint observations and completed collectives.

Assumption 3.1 (Analytic domain)**.** $f : \mathbb{R}^d \to \mathbb{R}$ has L-Lipschitz gradient, $L > 0$, and $f \geq f_* > -\infty$. All iterates and states are finite real quantities. Every relevant group has $\epsilon > 0$, $0 \leq \beta_1, \beta_2 < 1$, finite $\gamma, q \geq 0$, finite $p_{\text{ext}} \geq 0$, and $0 < c_{\max} \leq 1$. Local clocks are nonnegative integers; warmup and horizon settings are fixed finite integers. Initial moments and detector states are as in Section 2. There are no external parameter mutations.

These conditions are stronger than constructor acceptance. Epsilon and p_{ext} are not checked there, and arbitrary group overrides bypass some top-level checks. They are mathematical scope restrictions on supported inputs, not changes to executable code.

Assumption 3.2 (Canonical configuration for the main theorem)**.** Use Adam mode with AGC and centralization disabled, $\lambda = q = 0$, and all blocks active at every step. Groups share a constant learning rate $\alpha > 0$, fixed $\beta_1 = \beta$, fixed β_2, and a common $\epsilon > 0$. Detector settings may differ by block within Assumption 3.1; the exact implemented detector, warmup, cosine schedule, and auto-gamma are retained. Use ordinary initialized state, not an arbitrary resumed checkpoint. The supplied gradients satisfy $\|g_t\| \leq G$ for a finite deterministic G. There is no distributed execution requirement: the theorem is stated for a single process and the real recurrence of the scalar path.

The default positive weight decay is deliberately excluded. The main theorem is an actual selectable configuration of PsiLogic, not a claim about every constructor default. All proofs are for the real recurrence; fp32 state is not treated as exact real arithmetic. Equality of intended backend equations is separately qualified by tensor layout and precision.

Definition 3.3 (Oracle error and gap)**.** Set $\xi_t = g_t - h_t$, $\Delta = f(x_1) - f_*$, and $\Xi_T = \sum_{t=1}^{T} \|\xi_t\|^2$. The main pathwise theorem does not require unbiasedness or stochastic independence. Stochastic corollaries assume the stated second moments exist; x_1 is deterministic unless Δ is replaced by its expectation.

4 Preliminary bounds for the exact detector

Lemma 4.1 (Normalized EMA stability)**.** *For a nonempty block satisfying Assumption 3.1, $A_k \geq 0$ and $0 \leq Z_k \leq 100$ for $k \geq 2$. In the absence of DDP, $0 \leq F_k, S_k \leq 100$. These bounds also survive averaging nonnegative bounded rank states. If the postprocessed block gradient tends to zero and DDP is disabled, then $A_k, F_k, S_k \to 0$.*

Proof. The first seed has $A_1 = R_1 \geq 0$ and $F_1 = S_1 = 1$. Positivity of the EMA coefficients proves $A_k \geq 0$ by induction. For $k \geq 2$, $A_k + \epsilon \geq .01 R_k + \epsilon > 0$, so $0 \leq R_k/(A_k + \epsilon) \leq 100$. The updates of F, S are convex combinations of their old values and Z_k, so induction gives the interval bound. Arithmetic means preserve that interval.

For completeness, if $b \in [0,1)$ and $y_k = by_{k-1} + (1-b)a_k$ with $a_k \to 0$, then $y_k = b^{k-n}y_n + (1-b)\sum_{j=n+1}^{k} b^{k-j}a_j$. Given $\delta > 0$, choose n so $|a_j| \le \delta$ for $j > n$; the sum is at most δ in magnitude and the first term tends to zero. Thus $y_k \to 0$. Apply this argument to A when $R_k \to 0$. Since $Z_k \le R_k/\epsilon$, $Z_k \to 0$; applying it again to F, S proves the claim. □

Lemma 4.2 (Trust and effective-time bounds)**.** *Under Assumption 3.1, let* $c = \min_b[1 - \min(c_{\max}^{(b)}, .99)]$. *Then* $c \ge .01$ *and each active or inactive trust satisfies* $c \le u_t^{(b)} \le 1$. *For any nonnegative learning rates,*

$$c\sum_{t=1}^{T}\eta_t \le \sum_{t=1}^{T}\eta_t u_t^{(b)} \le \sum_{t=1}^{T}\eta_t.$$

The detector cannot by itself turn a nonsummable learning-rate sequence into a summable one.

Proof. The clipped cosine lies in $[-1,1]$, so $0 \le D \le 1$. The sigmoid lies in $(0,1)$ and $0 \le \tanh S \le 1$; hence $0 \le s \le 1$. The schedule and auto-gamma are nonnegative, and $w \in [0,1]$. Consequently the argument of the minimum in (6) is nonnegative, and the subtracted amount lies in $[0, c_*]$. Inactive trust is one. Multiplying $c \le u_t^{(b)} \le 1$ by $\eta_t \ge 0$ and summing proves the assertion. This says nothing about normalization, momentum direction, or stochastic bias. □

Lemma 4.3 (Bounds for the exact Adam scaling)**.** *Under Assumptions 3.1–3.2, write*

$$x_{t+1} = x_t - \alpha H_t \widehat{m}_t, \qquad (H_t)_{ii} = \frac{u_t^{(b(i))}\sqrt{1-\beta_2^t}}{\sqrt{v_{t,i}} + \epsilon}. \tag{11}$$

Then, in the Loewner order,

$$aI \preceq H_t \preceq MI, \qquad a = \frac{c}{G + \epsilon/\sqrt{1-\beta_2}} > 0, \qquad M = \epsilon^{-1}. \tag{12}$$

Also $\|\widehat{m}_t\| \le G$.

Proof. Expanding the variance recurrence gives $v_{t,i} = (1-\beta_2)\sum_{j=1}^{t}\beta_2^{t-j}g_{j,i}^2 \le G^2(1-\beta_2^t) \le G^2$, and it is nonnegative. Dividing numerator and denominator in (11) by $\sqrt{1-\beta_2^t}$ gives a denominator at most $G + \epsilon/\sqrt{1-\beta_2}$ and numerator at least c. This proves the lower bound. In the original form the numerator is at most one and the denominator at least ϵ, proving the upper bound; in particular $a \le M$. Expansion of the first moment gives

$$\widehat{m}_t = \sum_{j=1}^{t} w_{tj} g_j, \qquad w_{tj} = \frac{(1-\beta)\beta^{t-j}}{1-\beta^t}, \qquad w_{tj} \ge 0, \quad \sum_{j=1}^{t} w_{tj} = 1. \tag{13}$$

The triangle inequality gives the moment bound. For $\beta = 0$, these weights are understood as $w_{tt} = 1$ and $w_{tj} = 0$ for $j < t$. □

5 Level A: a simplified stochastic baseline

This section replaces momentum and adaptive variance by a predictable symmetric scaling B_t. For example, a scalar detector evaluated using only past gradients could supply such a multiplier. This is a *modified algorithm*; using the current implemented u_t does not satisfy its predictability assumption.

Theorem 5.1 (Predictable scaling)**.** *Suppose $x_{t+1} = x_t - \alpha B_t g_t$, with fixed constants $0 < a \le M$, $aI \preceq B_t \preceq MI$, B_t measurable with respect to $\mathcal{F}_{t-1}$, $\mathbb{E}[g_t \mid \mathcal{F}_{t-1}] = h_t$, and $\mathbb{E}[\|g_t - h_t\|^2 \mid \mathcal{F}_{t-1}] \le \sigma^2$. If f obeys Assumption 3.1 and $0 < \alpha \le a/(LM^2)$, then*

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\,\|h_t\|^2 \le \frac{2\Delta}{a\alpha T} + \frac{LM^2\alpha\sigma^2}{a}. \tag{14}$$

Choosing $\alpha = \min\{a/(LM^2), T^{-1/2}\}$ gives $O(T^{-1/2})$ for fixed problem constants.

Proof. Lipschitz continuity of the gradient implies the smoothness inequality $f(x+s) \le f(x) + \langle \nabla f(x), s\rangle + (L/2)\,\|s\|^2$: indeed integrate $\langle \nabla f(x+rs) - \nabla f(x), s\rangle$ from $r = 0$ to 1 and bound it by $\int_0^1 Lr\,\|s\|^2\,dr$. Apply it with $s = -\alpha B_t g_t$. These random variables are integrable at every finite time: from deterministic x_1, the bound $\|h_t\| \le \|\nabla f(0)\| + L\,\|x_t\|$ and the conditional variance hypothesis imply finite second moment of g_t whenever x_t has finite second moment. The recursion and $\|B_t\| \le M$ then imply finite second moment of x_{t+1}. Induction proves these facts, and $|f(x)| \le |f(0)| + \|\nabla f(0)\|\,\|x\| + (L/2)\,\|x\|^2$ implies objective integrability. Thus the conditional expectations below are defined. Predictability and conditional unbiasedness give $\mathbb{E}[\langle h_t, B_t g_t\rangle \mid \mathcal{F}_{t-1}] = \langle h_t, B_t h_t\rangle \ge a\,\|h_t\|^2$. Expansion of $g_t = h_t + (g_t - h_t)$ gives conditional second moment at most $\|h_t\|^2 + \sigma^2$, with zero cross term. Thus

$$\mathbb{E}[f(x_{t+1}) \mid \mathcal{F}_{t-1}] \le f(x_t) - \alpha(a - L\alpha M^2/2)\,\|h_t\|^2 + L\alpha^2 M^2\sigma^2/2.$$

The step condition leaves coefficient at least $a/2$. Take expectations, sum, and use $\mathbb{E}f(x_{T+1}) \ge f_*$. Division by $a\alpha T/2$ yields (14). The specified horizon-dependent step has the stated asymptotic order. No factorization involving a current-gradient scaling was used. □

6 Levels B–D: exact cancellation, moments, and bias correction

6.1 A pathwise momentum-tracking lemma

The following argument permits H_t to depend arbitrarily on the current gradient. This is the key distinction from Theorem 5.1. Define

$$K = \frac{\beta}{1-\beta}, \qquad C_\beta = \frac{1}{1-\beta}, \qquad e_t = \widehat{m}_t - h_t.$$

Lemma 6.1 (Finite-horizon tracking energy)**.** *For $g_t = h_t + \xi_t$, the weights in (13), and any sequence $h_1, \dots, h_T$, let $n_t = \sum_{j=1}^{t} w_{tj}\xi_j$ and $d_t = \sum_{j=1}^{t} w_{tj}h_j - h_t$. Then $e_t = n_t + d_t$ and*

$$\sum_{t=1}^{T} \|n_t\|^2 \le C_\beta \sum_{t=1}^{T} \|\xi_t\|^2, \tag{15}$$

$$\sum_{t=1}^{T} \|d_t\|^2 \le K^2 \sum_{t=2}^{T} \|h_t - h_{t-1}\|^2. \tag{16}$$

For the update (11) and an L-Lipschitz gradient, therefore,

$$\sum_{t=1}^{T} \|e_t\|^2 \le 2C_\beta \Xi_T + 2Q \sum_{t=1}^{T} \|\widehat{m}_t\|^2, \qquad Q = K^2L^2\alpha^2M^2. \tag{17}$$

Proof. The identity follows by expanding $g_j = h_j + \xi_j$. Convexity of squared norm (or Cauchy–Schwarz with weights summing to one) gives $\|n_t\|^2 \le \sum_{j\le t} w_{tj} \|\xi_j\|^2$. Since $1-\beta^t \ge 1-\beta$, $w_{tj} \le \beta^{t-j}$; hence $\sum_{t=j}^{T} w_{tj} \le \sum_{r=0}^{\infty} \beta^r = C_\beta$. Interchanging the two *finite* sums proves (15).

Let $b_t = \sum_{j\le t} w_{tj} h_j$. For $t \ge 2$, $b_t = \rho_t b_{t-1} + (1-\rho_t) h_t$, where

$$\rho_t = \frac{\beta(1-\beta^{t-1})}{1-\beta^t} \in [0, \beta].$$

Thus $d_1 = 0$ and $d_t = \rho_t(d_{t-1} + h_{t-1} - h_t)$. Iterating its norm inequality gives $\|d_t\| \le \sum_{j=2}^{t} \beta^{t-j+1} \|h_{j-1} - h_j\|$. For $\beta > 0$, weighted Cauchy–Schwarz bounds its square by

$$\left(\sum_{j=2}^{t} \beta^{t-j+1}\right)\left(\sum_{j=2}^{t} \beta^{t-j+1} \|h_{j-1} - h_j\|^2\right).$$

The first factor is at most K; summing the second factors over t gives at most $K \sum_{j=2}^{T} \|h_{j-1} - h_j\|^2$. This proves (16). If $\beta = 0$, $d_t = 0$ and the result holds directly. Finally $\|n_t + d_t\|^2 \le 2\|n_t\|^2 + 2\|d_t\|^2$, while

$$\|h_t - h_{t-1}\| \le L \|x_t - x_{t-1}\| \le L\alpha M \|\widehat{m}_{t-1}\|.$$

The last sum ranges over $t-1 = 1, \dots, T-1$ and can be enlarged to $1, \dots, T$, since all terms are nonnegative. This proves (17). The zero initial drift and the first oracle error $n_1 = \xi_1$ have both been retained. □

6.2 Main theorem and proof

Theorem 6.2 (Canonical Adam configuration: pathwise bound)**.** *Under Assumptions 3.1–3.2, use a, M from (12), K, C_β above, and the constant step*

$$0 < \alpha \le \frac{a}{4LM^2 \max\{1, K\}}. \tag{18}$$

Then every finite trajectory of the exact real recurrence satisfies

$$\sum_{t=1}^{T} \|\nabla f(x_t)\|^2 \le \frac{9\Delta}{\alpha a} + 13 C_\beta \frac{M^2}{a^2} \sum_{t=1}^{T} \|g_t - \nabla f(x_t)\|^2. \tag{19}$$

The result permits all dependencies introduced by the exact detector and second-moment normalization. It assumes no independence, no monotonicity of H_t, and no relation such as $\beta_1^2 < \beta_2$.

Proof. Write $z_t = \widehat{m}_t$ for this proof, so $h_t = z_t - e_t$. Using $aI \preceq H_t \preceq MI$, Cauchy–Schwarz, and $M\|e_t\|\|z_t\| \le (a/2)\|z_t\|^2 + (M^2/(2a))\|e_t\|^2$,

$$\langle h_t, H_t z_t \rangle = \langle z_t, H_t z_t \rangle - \langle e_t, H_t z_t \rangle \ge \frac{a}{2}\|z_t\|^2 - \frac{M^2}{2a}\|e_t\|^2.$$

The smoothness inequality proved in Theorem 5.1 gives

$$f(x_{t+1}) \le f(x_t) - \alpha\left(\frac{a}{2} - \frac{L\alpha M^2}{2}\right)\|z_t\|^2 + \frac{\alpha M^2}{2a}\|e_t\|^2. \tag{20}$$

Let $U = \sum_{t=1}^{T} \|z_t\|^2$. Summing (20), using $f(x_{T+1}) \geq f_*$ and (17), yields

$$\alpha \left(\frac{a}{2} - \frac{L\alpha M^2}{2} - \frac{M^2 Q}{a} \right) U \leq \Delta + \frac{\alpha M^2 C_\beta}{a} \Xi_T. \tag{21}$$

Condition (18) implies $L\alpha M^2/2 \leq a/8$ and

$$Q = K^2 L^2 \alpha^2 M^2 \leq \frac{a^2}{16M^2} \leq \frac{1}{16}.$$

Consequently $M^2 Q/a \leq a/16$, so the coefficient in parentheses is at least $5a/16 \geq a/4$. Therefore

$$U \leq \frac{4\Delta}{\alpha a} + \frac{4M^2 C_\beta}{a^2} \Xi_T. \tag{22}$$

Because $h_t = z_t - e_t$,

$$\sum_{t=1}^{T} \|h_t\|^2 \leq 2U + 2\sum_{t} \|e_t\|^2 \leq (2 + 4Q)U + 4C_\beta \Xi_T \leq \frac{9}{4} U + 4C_\beta \Xi_T.$$

Substitution of (22) gives coefficient 9 on $\Delta/(\alpha a)$ and coefficient $C_\beta(9M^2/a^2 + 4) \leq 13 C_\beta M^2/a^2$ on Ξ_T, since $a \leq M$. This proves (19). Every inequality was pathwise before any expectation was taken. □

Corollary 6.3 (Deterministic stationarity)**.** *If $g_t = \nabla f(x_t)$ in Theorem 6.2, then*

$$\frac{1}{T} \sum_{t=1}^{T} \|\nabla f(x_t)\|^2 \leq \frac{9\Delta}{\alpha a T}.$$

For an infinite trajectory satisfying its assumptions, $\|\nabla f(x_t)\| \to 0$. This does not assert that x_t converges or that its limit, if any, is a local minimum.

Proof. Set $\Xi_T = 0$. The partial sums of nonnegative squared gradients are bounded uniformly in T; hence their series converges. If its terms did not tend to zero, infinitely many would exceed one fixed positive number, contradicting boundedness of the partial sums. □

Corollary 6.4 (Stochastic error and growing batches)**.** *If the pathwise assumptions of Theorem 6.2 hold almost surely and $\mathbb{E}\,\|\xi_t\|^2 \leq \sigma_t^2$, then*

$$\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}\,\|h_t\|^2 \leq \frac{9\Delta}{\alpha a T} + 13 C_\beta \frac{M^2}{a^2} \frac{1}{T} \sum_{t=1}^{T} \sigma_t^2. \tag{23}$$

If $\sum_t \sigma_t^2 < \infty$, the gradients tend to zero both in mean square and almost surely. If only their Cesàro mean tends to zero, the displayed average tends to zero; this alone is not a last-iterate assertion.

In particular, suppose each g_t is the average of B conditionally independent unbiased samples, each with conditional variance at most σ^2, and the averages satisfy the required bound G. For a run of T iterations choose $B = \lceil \sqrt{T} \rceil$. Then (23) is $O(T^{-1/2})$ at $T\lceil \sqrt{T} \rceil$ sample evaluations. This is not a single-sample $O(T^{-1/2})$ theorem.

Proof. Take expectations of the finite pathwise inequality; no limit interchange is needed for (23). If the error-variance series is finite, the expected partial sums of $\|\xi_t\|^2$ are uniformly bounded. Monotone convergence shows that their infinite sum has finite expectation and is therefore finite almost surely. The pathwise theorem then bounds the infinite gradient-energy series almost surely, implying pointwise convergence to zero as in Corollary 6.3. Taking expectations in (19) bounds the series of expected squared gradients, which likewise has terms tending to zero. For the batch assertion, expanding the squared norm of the average noise gives B diagonal contributions at most σ^2 and zero cross terms by conditional independence and zero means. Division by B^2 gives $\sigma_t^2 \le \sigma^2/B$, and substitution proves the rate and sample count. □

Corollary 6.5 (Relative oracle error)**.** *Suppose instead that* $\mathbb{E}[\|\xi_t\|^2 \mid \mathcal{F}_{t-1}] \le r^2 \|h_t\|^2$ *and* $\vartheta = 13C_\beta(M/a)^2r^2 < 1$*, with all assumptions of Theorem 6.2 holding almost surely. Then*

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\,\|h_t\|^2 \le \frac{9\Delta}{(1-\vartheta)\alpha a T},$$

and gradients converge to zero in mean square and almost surely.

Proof. Take expectations in (19) and apply the conditional error bound. Move $\vartheta\sum_t \mathbb{E}\,\|h_t\|^2$ to the left and divide by $1-\vartheta > 0$. The resulting bound on all finite partial sums implies summability of expected squared gradients. Monotone convergence implies almost-sure summability too, and the same nonnegative-series argument proves both assertions. Required finite moments follow, for each finite horizon, from $\|g_t\| \le G$, $\|\widehat{m}_t\| \le G$, and $\|x_t\| \le \|x_1\| + (t-1)\alpha MG$, together with Lipschitz continuity of the gradient. □

Remark 6.6 (What persistent noise permits here). For a fixed nonzero variance bound, (23) has a nonvanishing term $13C_\beta(M/a)^2\sigma^2$. Taking a smaller constant step does not remove this term. This is an upper bound, not a proof of a matching error floor, but it cannot be cited as convergence to stationarity. The very small relative-error condition is a sufficient hypothesis, not a property implied by unbiasedness or the implementation.

7 Stronger and specialized results

Proposition 7.1 (Sharper no-momentum deterministic bound)**.** *Under Assumptions 3.1–3.2, with* $\beta_1 = 0$ *and* $g_t = h_t$*, any* $0 < \alpha \le a/(LM^2)$ *gives*

$$f(x_{t+1}) \le f(x_t) - \frac{\alpha a}{2}\|h_t\|^2, \qquad \frac{1}{T}\sum_{t=1}^{T}\|h_t\|^2 \le \frac{2\Delta}{\alpha a T}.$$

If, along the trajectory, $\|\nabla f(x)\|^2 \ge 2\mu(f(x) - f_*)$ *for* $\mu > 0$ *and additionally* $\alpha a\mu \le 1$*, then* $f(x_{T+1}) - f_* \le (1-\alpha a\mu)^T\Delta$.

Proof. Now $\widehat{m}_t = h_t$. Smoothness and the scaling bounds give $f(x_{t+1}) \le f(x_t) - \alpha a\|h_t\|^2 + (L\alpha^2M^2/2)\|h_t\|^2$. The step condition proves the one-step claim, and summing proves the average bound. Inserting the stated gradient dominance inequality gives a nonnegative contraction factor $1-\alpha a\mu$ at each step; iterating that inequality T times proves the last assertion. □

The gradient dominance condition holds for a differentiable μ-strongly convex function with attained minimum, taking $f_* = f(x_*) = \min f$: strong convexity gives $f_* \ge f(x) + \langle \nabla f(x), x_* - x\rangle +$

$(\mu/2)\|x_* - x\|^2 \geq f(x) - \|\nabla f(x)\|^2/(2\mu)$ by completing the square. The bounded-gradient hypothesis here is along the trajectory, so it does not assert globally bounded gradients for a strongly convex function.

Proposition 7.2 (A convex consequence of the full-momentum theorem)**.** *Under Corollary 6.3, suppose f is convex, has minimizer x_*, $f_* = f(x_*)$, and $\|x_t - x_*\| \leq R$ for all t. For $\bar{x}_T = T^{-1}\sum_{t=1}^{T} x_t$,*

$$f(\bar{x}_T) - f_* \leq 3R\sqrt{\frac{\Delta}{\alpha a T}}.$$

Proof. Convexity gives $f(x_t) - f_* \leq \langle h_t, x_t - x_*\rangle \leq R\|h_t\|$. Jensen's inequality for the convex function f and Cauchy–Schwarz for the finite sum give $f(\bar{x}_T) - f_* \leq RT^{-1}\sum_t \|h_t\| \leq R\sqrt{T^{-1}\sum_t \|h_t\|^2}$. Apply Corollary 6.3 and take the square root. The bounded-distance assumption is explicit; it was not derived from smoothness. □

Proposition 7.3 (Single-coordinate adaptive-step bound)**.** *For the Adam adaptive term in* (8)*, with $\beta_2 > \beta_1^2$, $\beta_1, \beta_2 \in [0, 1)$, $\epsilon > 0$, $\eta_t > 0$, zero initial moments, $0 \leq u_t \leq 1$, and arbitrary finite gradients, its magnitude divided by η_t in each coordinate is at most*

$$\frac{(1-\beta_1)\sqrt{1-\beta_2^t}}{(1-\beta_1^t)\sqrt{1-\beta_2}}\sqrt{\sum_{j=0}^{t-1}(\beta_1^2/\beta_2)^j} \leq \frac{1}{\sqrt{(1-\beta_2)(1-\beta_1^2/\beta_2)}}. \tag{24}$$

This bounds the adaptive step, not the decay contributions or generalization error, and does not prove an advantage over AdamW.

Proof. Cauchy–Schwarz applied to $\sum_{j=0}^{t-1}\beta_1^j g_{t-j,i} = \sum_{j=0}^{t-1}(\beta_1/\sqrt{\beta_2})^j(\sqrt{\beta_2})^j g_{t-j,i}$ gives

$$m_{t,i}^2 \leq \frac{(1-\beta_1)^2}{1-\beta_2}\left(\sum_{j=0}^{t-1}(\beta_1^2/\beta_2)^j\right) v_{t,i}.$$

If $v_{t,i} = 0$, every corresponding gradient and $m_{t,i}$ are zero. Otherwise divide by $(\sqrt{v_{t,i}} + \epsilon)^2 \geq v_{t,i}$, multiply by $u_t^2(1-\beta_2^t)/(1-\beta_1^t)^2$, and take square roots. The geometric sum is at most $1/(1-\beta_1^2/\beta_2)$; $(1-\beta_1)/(1-\beta_1^t) \leq 1$ and $\sqrt{1-\beta_2^t} \leq 1$ give the second bound. No probability statement is involved. □

Sensitivity and limits of the constants. The cancellation cap enters through c; stronger allowed cancellation reduces a and worsens these worst-case constants. This is not evidence that actual trajectories worsen. EMA coefficients .9 and .99 determine responsiveness and the bound 100; the main theorem only needs the resulting trust interval. For a constant nonzero gradient norm R, $A_k = R$ and $Z_k = R/(R+\epsilon)$ for $k \geq 2$, so F_k, S_k tend to this nonzero value. Thus small absolute norms do not generally imply a small normalized detector before the epsilon scale is reached. Lemma 4.1 supplies the correct asymptotic assertion. There is no online-regret theorem or proven improvement in training efficiency in this work.

For scale, setting $L = G = 1$, $\beta_1 = .9$, $\beta_2 = .999$, $\epsilon = 10^{-8}$ and $c = .95$ gives a sufficient step upper bound of about 2.64×10^{-18} in (18). This worst-case estimate does not certify an ordinary training step such as 10^{-3}. Its conservatism is a substantial practical limitation, not a claimed necessary stability threshold.

8 Counterexamples and limitations

Proposition 8.1 (A fixed-step two-cycle in a legal configuration)**.** *Let $f(x) = x^2/2$, $\beta_1 = \beta_2 = 0$, $\gamma = \lambda = q = 0$, no preprocessing, and $\epsilon > 0$. For constant $\alpha > 2\epsilon$ and $x_1 = \alpha/2 - \epsilon > 0$, the real recurrence has $x_{t+1} = -x_t$. It does not converge to stationarity.*

Proof. The exact step is $x^+ = x - \alpha x/(|x| + \epsilon)$. At $|x| = \alpha/2 - \epsilon$ the denominator is $\alpha/2$, so $x^+ = -x$. Induction gives the cycle, with constant nonzero squared gradient $(\alpha/2 - \epsilon)^2$. This is a smooth strongly convex lower-bounded objective and a legitimate configuration. It does not contradict the small step requirement in Theorem 6.2. □

Proposition 8.2 (Current-sample normalization can reverse expected descent)**.** *At $x = 0$ for $f(x) = (x + .1)^2/2$, use an unbiased stochastic gradient equal to 10 with probability .1 and -1 with probability .9. In the first Adam step with $\beta_1 = \beta_2 = 0$, $\epsilon = .01$, and $\lambda = q = 0$, the expected update direction points uphill, both with inactive chaos and with the active first-step detector using fixed admissible settings. In the latter case let its common trust be $u > 0$. Then every $\alpha > 0$ increases expected objective in this first step.*

Proof. The gradient mean is $1 - .9 = .1 = f'(0)$; the oracle can be extended to every x as $g(x) = x + Z$ with the displayed two-point Z, and its noise has finite constant variance. At the first step $F = S = 1$ and $D = 0$, so the detector trust is independent of the gradient magnitude and has the same value u in both outcomes. The mean normalized direction is

$$\mathbb{E}\left[\frac{ug}{|g| + .01}\right] = u\left(\frac{100}{1001} - \frac{90}{101}\right) < 0.$$

Writing $V = ug/(|g| + .01)$, direct expansion of the quadratic gives $\mathbb{E}f(-\alpha V) - f(0) = -.1\alpha\mathbb{E}V + (\alpha^2/2)\mathbb{E}V^2 > 0$. This disproves a universal one-step expected-descent inequality under unbiasedness alone. A single uphill step is not a proof of asymptotic nonconvergence, and we make no such inference from this example. □

Proposition 8.3 (Bounded positive cancellation alone is insufficient)**.** *For the same two-point gradient, a current-gradient multiplier in $[.01, 1]$ can have $\mathbb{E}[ug] < 0$ despite $\mathbb{E}g > 0$.*

Proof. Take $u(10) = .01$ and $u(-1) = 1$. Then $\mathbb{E}[ug] = .01 - .9 = -.89$ while $\mathbb{E}g = .1$. This generic multiplier example rules out a proof based only on its interval bounds. It is not asserted to be the exact implemented detector; a reachable-history calculation for that detector is recorded separately in the following proposition and `counterexamples.md`. □

Proposition 8.4 (Anti-alignment from the exact detector)**.** *Set $\beta_1 = \beta_2 = 0$, $\epsilon = .01$, $\gamma = p_{\text{ext}} = 1$, $c_{\max} = 1$, zero warmup and horizon, adaptive thresholding with $\tau_{\text{scale}} = 2$, auto-gamma off, and no decay or preprocessing. There is a reachable history and an unbiased next gradient with positive conditional mean for which the implemented trust satisfies $\mathbb{E}[u_2 g_2 \mid \mathcal{F}_1] < -463/2000$.*

Proof. After a first gradient -1, the old momentum is -1 and $A_1 = F_1 = S_1 = 1$. At step two take $g = 10$ with probability $1/10$ and $g = -1$ otherwise. Its conditional mean is $1/10$. For $g = 10$, direct substitution gives

$$A_+ = 109/100, \quad Z_+ = 100/11, \quad F_+ = 199/110,$$
$$S_+ = 1189/1100, \quad F_+/(S_+ + \epsilon) = 199/120, \quad D_+ = 2001/2002 > 99/100.$$

Thus the sigmoid excess exceeds $1/2$, and $s_+ > (3/4)(1/2)(1/2 + 99/100) = 447/800$, since $S_+ > 1$ and $\tanh(1) > 3/4$. The latter inequality follows from $\exp(1) > 1+1+1/2+1/6 = 8/3$ and therefore $\exp(2) > 7$. The cap .99 also exceeds $447/800$, so $u_+ < 353/800$. For $g = -1$ the values are

$$A_- = 1, \quad F_- = 1009/1010, \quad S_- = 10099/10100,$$
$$F_-/(S_- + \epsilon) = 1009/1020 < 1, \quad D_- = 1/202 < 1/200.$$

The excess is less than $1/2$ and $\tanh S_- < 1$; hence $s_- < 101/400$ and $u_- > 299/400$. It follows that

$$\mathbb{E}[ug \mid \mathcal{F}_1] = u_+ - (9/10)u_- < 353/800 - (9/10)(299/400) = -463/2000.$$

To realize this history with an everywhere unbiased oracle for a smooth lower-bounded objective, fix $\alpha > 0$ and start at zero. Let $u_1 > 0$ be its deterministic first-step trust and $\delta = \alpha u_1/(1+\epsilon) > 0$. Set $L_0 = 11/(10\delta)$ and $f(x) = L_0(x - 1/L_0)^2/2$. Then $f'(0) = -1$ and $f'(\delta) = 1/10$. Use the exact gradient except at δ, where the displayed two-point oracle is used. This oracle is unbiased everywhere, has variance at most $1089/100$, and reaches the required history with probability one. Its definition need not be continuous in x; no such continuity was assumed for an unbiased stochastic oracle. This is a finite-step obstruction, not an asymptotic nonconvergence claim. □

Proposition 8.5 (Gradient centralization can stall permanently)**.** *For a 1×2 parameter tensor X, let $f(X) = \|X - (1,1)\|^2/2$ and $X_1 = (0,0)$. With gradient centralization on and initialized zero moments, the implemented real recurrence stays at zero for any learning-rate schedule, in either Adam or Lion mode, even with AGC and either decay enabled, provided its arithmetic is well-defined. The true squared gradient remains 2.*

Proof. At zero the raw gradient is $(-1,-1)$. Whole-tensor AGC multiplies both coordinates by the same scalar, and centralization subtracts their common mean, giving $g = 0$. Both moment states remain zero and the Adam/Lion adaptive update is zero. Both decay operations multiply the zero parameter, preserving it. Induction repeats the argument at every step. Meanwhile $\nabla f(0) = (-1,-1)$, of squared norm 2. Thus projected-gradient reasoning cannot silently be used to claim stationarity of the original objective. □

AGC, decay, and accepted but invalid inputs. Clipping the two-point oracle above to magnitude one gives mean $.1 - .9 = -.8$, reversing its original mean. Whole-tensor AGC realizes this at a parameter norm and AGC coefficient whose threshold in (1) equals one. Unbiasedness must therefore be imposed on the *postprocessed* oracle or its bias must be controlled explicitly. Under the standard unclipped oracle assumption this is not automatic.

For $f(x) = (x-1)^2/2$ at $x = 1$ with zero initial moments, weight decay $\lambda > 0$ and $\alpha\lambda \neq 0$ moves a stationary point to $1 - \alpha\lambda$. In general the no-quantum-decay direction is $H_t\widehat{m}_t + \lambda x_t$, not $H_t(\widehat{m}_t + \lambda x_t)$. When H_t varies with time and history, this is not the gradient step for a fixed objective $f + \lambda\|x\|^2/2$. Quantum decay further adds a current-gradient-dependent coordinate shrinkage, with the cross term in (8). Neither is covered by the main theorem. A conditional theorem for a specially chosen fixed metric would not establish the default case.

Negative p_{ext}, which is not rejected by the constructor, can produce $u > 1$ because the clamp has only an upper bound. Accepted $\epsilon = 0$ with zero gradient and variance makes the Adam quotient $0/0$; nonfinite hyperparameters or data cause further failures. Large $\eta\lambda$ or ηQ_k can change the sign of a decay factor. Such decay is outside the main theorem's zero-decay configuration. Invalid epsilon, negative p_{ext}, and nonfinite inputs are outside the analytic domain; large positive decay products can be legal finite inputs, but the decay factors need not be contractions.

Finite precision. Even finite positive epsilon and bounded trust do not prevent floating-point stalling, overflow, underflow, or cancellation. If a nonzero real increment is smaller than half an ulp at a stored parameter and rounding returns the old value, the real update and stored trajectory differ. No summable rounding error bound is established for this implementation. The real-arithmetic stationarity theorem therefore does not imply exact convergence of fp16, bfloat16, fp32, or fp64 program executions.

9 Relationship to adaptive optimization theory

Adam introduced adaptive first and second moments with bias correction [4]. Its original convergence arguments cannot be used as an unconditional inheritance theorem: Reddi et al. [6] exhibit failure and motivate AMSGrad's long-term second-moment memory. PsiLogic's current variance has no running maximum, so it is not AMSGrad. Adding one would be an algorithmic modification and would still require an analysis of the extra current-gradient multiplier.

Chen et al. [2] give sufficient conditions for classes of Adam-type methods and analyze effective-step behavior. We do not assert those conditions are implied by this detector. Theorem 6.2 instead controls the momentum tracking error pathwise under a uniformly positive bounded scaling and a small constant step. Its deterministic and vanishing-error assumptions explain why it is compatible with known stochastic failures.

Decoupled weight decay differs from loss-based L_2 regularization for adaptive methods [5]; hence our stationarity target and decay exclusions are essential. The Lion branch follows a different sign-momentum structure [1] and is specified but not covered by the Adam proof. The simplified stochastic benchmark connects to smooth nonconvex stationarity bounds [3]. Matching its iteration rate under stronger conditions or larger batches is not evidence of improved oracle complexity, robustness, or training efficiency.

10 Implementation-to-mathematics correspondence

The authoritative import/package root is `psilogic/`. Public constructors, presets, convenience selection, integrations, and group builders ultimately construct `PsiLogic`; they do not introduce another optimizer update implementation. The generated `build/lib/psilogic` copy has stale CUDA files and is separately recorded, not silently substituted for the active source. Untracked paper scripts do not alter the update path. The full trace and defaults are in `implementation_trace.md` and `algorithm_specification.md`; detailed kernel scope is in `backend_trace.md`.

Mathematical object	Source location in inspected working tree
AGC, centralization, initial state	`optimizer.py:96--126`
Defaults, parameter options	`optimizer.py:283--344`
Schedules and warmup	`_chaos.py:35--61,114--124`
Disagreement and soft signal	`_chaos.py:127--186`
Trust and hard cap	`_chaos.py:186--207`
Norm EMA initialization/update	`_chaos.py:263--282`
Sequential decay, active branching	`optimizer.py:472--527`
Adam/Lion update and epsilon	`optimizer.py:529--558`
Scalar preparation and old momentum	`optimizer.py:560--678`
Foreach preparation/finalization	`optimizer.py:680--917`
Dispatch	`optimizer.py:920--974`
Fused orchestration	`_cuda/step.py:320--668`
Fused Adam/Lion arithmetic	`_cuda/kernels.py:239--373`

Paths in the table are relative to `psilogic/`. References denote the snapshot, not a promise that future edits retain line numbers.

The scalar branch is used on CPU and when compiling; ordinary CUDA dispatch prefers fused kernels, then foreach. On finite well-defined inputs, the intended real recurrences agree for supported dense contiguous CUDA layouts with fresh gradients each call. This statement is deliberately narrower than bitwise equivalence: moments are normally fp32; low-precision norm arithmetic, fused reductions, host versus tensor auto-gamma, and intermediate parameter rounding differ. Fused quantum decay evaluates a sigmoid identity for tanh. Fused flat pointer indexing does not validate parameter contiguity; mixed CPU/CUDA groups can reach an unsuitable launch. Foreach and homogeneous fused AGC mutate gradient buffers in place, unlike the scalar path. Repeated steps without replacing gradients can therefore differ even in the real model. DDP averages fast/slow only and does not guarantee identical trust for ranks with different local gradients or momentum. Resumed state may have different initial values or dtypes and needs separate hypotheses.

11 Formal verification status

Ten declarations in `lean/PsiLogic.lean` were successfully compiled and kernel-checked with Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The proof-check command exited with status zero. The raw output, source hash, command, and toolchain are in `lean/build.log` and `lean/build_status.json`.

The checked statements cover the cap-dependent trust interval and .01 floor; the normalized-EMA upper bound and preservation of the interval $[0, 100]$; scalar identities for epsilon bias correction and denominator scaling; absorption of a scalar smoothness-error term; finite telescoping; a weighted certificate from a terminal objective lower bound; and the corresponding finite average $1/T$ certificate. The last four statements assume the relevant scalar descent inequality explicitly. They do not derive it from differentiability, the vector optimizer, or a stochastic process. All ten axiom reports list only `propext`, `Classical.choice`, and `Quot.sound`; there are no placeholder proofs or added axioms.

This is meaningful verification of a limited algebraic core, not a formal proof of Theorem 6.2, the stochastic corollaries, Python execution semantics, or Triton memory-layout correctness. Remaining formalization work includes the normed-space smoothness argument, vector momentum convolution bounds, the connection to the actual recurrence, and conditional-expectation and infinite-series results. Neither numerical parity nor repeated mathematical review substitutes for these missing formal developments.

12 Reproducibility, audit, and theorem status

The checks in this section describe the original investigation snapshot. AI agents prepared the source inspection, proof audit, and second derivation; these checks are not external peer review. Later publication edits to comments and documentation are recorded separately from the original source hashes.

Before proof construction, an independent double-precision recurrence was compared with actual CPU scalar and explicitly invoked CPU foreach execution. The check passed 34 cases and 47,380 parameter/state comparisons at absolute and relative tolerance 2×10^{-5}, including Adam, Lion, AGC followed by centralization, quantum decay, warmup, cosine schedules, auto-gamma, missing gradients, unequal counters, zero-norm initialization, and the epsilon placement. These are numerical correspondence tests, not proofs of exact arithmetic equivalence. GPU execution was unavailable. The source hashes and results are retained with the scripts.

The selected existing regression suite passed 72 tests, with 16 CUDA-dependent tests skipped. Separate checks verified exact symbolic identities and tested 300 finite momentum-tracking sequences and eight synthetic trajectories against the proof inequalities. Separate checks reproduced the analytic counterexamples. At the time of the investigation, SHA-256 comparisons confirmed that all 14 inspected package source files were unchanged. Build commands and results accompany the artifacts.

The proof audit tracks conditional measurability, initial-state terms, coefficient signs, index shifts, and the distinction between pathwise and expected statements. In particular: the current detector is never treated as predictable; the first oracle error is not dropped; the momentum drift sum ends at $T-1$ before being enlarged; the step bound is used before error absorption; and only summable nonnegative series justify last-iterate conclusions. A second derivation and algebraic stress checks are recorded separately; neither is represented as formal verification.

Result	Scope/status of mathematical claim
Detector/EMA and trust bounds	Complete real-arithmetic proofs; explicit positive-domain assumptions
Predictable stochastic $O(T^{-1/2})$	Complete proof for modified simplified method only
Full-momentum bound (19)	Complete pathwise proof for Assumption 3.2
Deterministic $O(T^{-1})$ and stationarity	Consequences under bounded gradients and small constant step
Stochastic vanishing/relative error	Conditional consequences; batch/sample cost stated
No-momentum gradient dominance	Conditional linear objective convergence
Counterexamples	Exact analytic constructions; one-step and asymptotic claims distinguished
Default decay, arbitrary fixed noise, Lion	No general stationarity theorem established
Floating point, GPU/DDP semantics	Not formally verified; source-level limitations disclosed
Lean scalar and finite-sum core	Ten declarations kernel-checked; full convergence theorem not formalized

Remaining mathematical limitations are substantive. There is no theorem for the full default configuration with positive weight decay, arbitrary persistent unbiased noise, centralization, quantum decay, or general Lion updates; no guarantee for arbitrary group clocks or loaded checkpoints; no proof that practically used learning rates meet the conservative sufficient bound; and no theorem of superiority over AdamW. These exclusions are visible assumptions and open obligations, not hidden gaps in a claimed broader result. The separate proof-gap register distinguishes rejected claims from

proved conditional results. The original investigation left the optimizer source and pre-existing research artifacts unchanged. It did not publish or submit them.

References

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